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Department of Information Technology

Software Requirements Specification (SRS)

Project Title: Vehicle Rental Booking Portal

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1. Introduction

1.1 Problem Statement

Vehicle rental services are becoming increasingly popular, but many rental businesses still rely on manual processes or outdated systems to manage bookings, customer records, and payments. This often leads to delayed bookings, duplicate reservations, inaccurate vehicle availability, and errors in maintaining records.

These issues affect both customers and rental agencies. Customers may experience delays in booking or confirmation, while administrators spend considerable time managing vehicles, payments, and rental records manually.

To overcome these challenges, a centralized Vehicle Rental Booking Platform is proposed. The system will automate the rental process, improve accuracy, reduce manual work, and provide a better experience for both customers and administrators.

1.2 Need for the Project

As online services continue to grow, customers expect a quick and convenient way to rent vehicles. Manual booking systems are no longer efficient because they are time-consuming, prone to errors, and difficult to manage as the number of customers increases.

This project provides a single platform to manage vehicle bookings, customer information, payments, and rental records. It aims to simplify daily operations, improve data accuracy, prevent duplicate bookings, and offer a secure and user-friendly experience for both customers and administrators

1.3 Objectives

The main objectives of the Vehicle Rental Booking Platform are:

- To provide a secure online platform for vehicle booking and rental management.
- To automate vehicle availability tracking and prevent duplicate bookings.
- To reduce manual work by managing customer, booking, and payment records digitally.
- To implement secure authentication and role-based access for users.
- To generate invoices, reports, and business analytics automatically.
- To improve the overall efficiency and user experience of the rental process.

1.4 Scope

In Scope

The Vehicle Rental Booking Platform will allow customers to register, log in, search available vehicles, make bookings, complete payments, and view their booking history. Administrators will be able to manage vehicles, customers, bookings, payments, invoices, and reports through a centralized dashboard. The system will also maintain vehicle availability and support secure authentication with role-based access.

Out of Scope

The current version of the system will not include:

- Live GPS vehicle tracking
- Driver or chauffeur management
- Vehicle maintenance management
- Mobile applications (Android/iOS)
- Integration with external ERP or accounting systems
- AI-based pricing or demand prediction

2. Problem Identification and Scope

2.1 Existing System

Current Process

Many vehicle rental businesses still manage bookings, customer details, and payments manually or using basic software. Customers often need to contact the rental agency to check vehicle availability and confirm bookings, making the process slow and less efficient.

Limitations

- No real-time vehicle availability.
- Manual record management is time-consuming.
- Higher chances of booking errors and duplicate reservations.
- Report generation is difficult and requires extra effort.
- Limited convenience for customers.

Problems Faced

The existing system results in delayed bookings, inaccurate records, duplicate reservations, and increased administrative workload, affecting both customers and rental agencies.

2.2 Proposed System

The proposed Vehicle Rental Booking Platform is a web-based application that automates the complete rental process. It allows customers to register, search vehicles, make bookings, complete payments, and view booking history. Administrators can manage vehicles, customers, bookings, payments, and reports through a centralized dashboard. The system also supports multi-branch management, enabling rental businesses to manage vehicles and operations across multiple branch locations from a single platform.

Advantages

- Automates the complete rental process.
- Provides real-time vehicle availability.
- Reduces manual work and booking errors.
- Prevents duplicate reservations.
- Generates invoices and reports automatically.
- Supports centralized management of multiple rental branches.
- Improves operational efficiency and record management.

Innovation

The platform integrates vehicle management, booking management, payment processing, invoice generation, and multi-branch operations into a single system. It also incorporates concurrent booking management to prevent reservation conflicts and ensure accurate vehicle availability across all branches.

2.3 Expected Benefits

- **Time Saving:** Faster booking and record management.
- **Cost Reduction:** Reduces paperwork and manual effort.
- **Improved Accuracy:** Minimizes errors in bookings and payments.
- **User Convenience:** Enables customers to book vehicles online anytime.
- **Better Management:** Provides administrators with centralized control and reporting.

3. Feasibility Study

3.1 Technical Feasibility

The proposed Vehicle Rental Booking Platform is technically feasible as it uses modern and widely adopted web development technologies. The frontend will be developed using React.js with Vite and Tailwind CSS, while the backend will be built using Node.js and Express.js. PostgreSQL will be used as the database, and JWT with bcrypt will provide secure user authentication. The project follows a REST API architecture and standard software engineering practices.

Software Tools

- Visual Studio Code
- pgadmin
- Postman
- Git & GitHub
- dbdiagram.io & plantuml (visual paradigm) (Documentation)

Hardware Requirements for Admin

- **Processor:** Intel Core i3 / AMD Ryzen 3 or higher
- **RAM:** 8 GB minimum
- **Storage:** 5 GB free space
- **Operating System:** Windows, macOS, or Linux
- **Internet Connection:** Required for development and deployment

3.2 Economic Feasibility

The project is economically feasible because it mainly uses free and open-source technologies. Software such as React.js, Node.js, Express.js, PostgreSQL, Git, GitHub, Visual Studio Code, and Postman can be used without licensing costs. Since no specialized hardware is required, the overall development cost is minimal.

3.3 Operational Feasibility

The system is designed with a simple and user-friendly interface, making it easy for both customers and administrators to use. Basic computer knowledge is sufficient to operate the application, and only minimal training is required for administrators to manage vehicles, bookings, payments, and reports.

3.4 Project Scope

Major Modules

- User Registration and Login
- Vehicle Management
- Vehicle Search and Booking
- Payment Processing
- Booking and Rental Management
- Invoice Generation
- Admin Dashboard
- Reports and Analytics

Deliverables

- Responsive Web Application
- PostgreSQL Database
- REST API
- Project Documentation (SRS, UML Diagrams)
- Testing Report
- Source Code and GitHub Repository

Timeline

The project will be completed in phases, including requirement analysis, system design, development, testing, documentation, and final deployment. Each phase will be completed according to the project schedule to ensure timely delivery.

4. 1 Stakeholder Analysis

Stakeholder	Role	Expectations
Adminstrator	Responsible for managing vehicle categories, vehicle records, customer accounts, bookings, payments, invoices, maintenance records, reports, and branch operations. The administrator monitors vehicle availability, updates rental charges, handles booking requests, tracks transactions, and oversees overall system performance.	A centralized platform for managing all rental operations, real-time visibility of vehicle availability and bookings, accurate payment and invoice records, automated report generation, efficient branch management, and reliable business insights for decision-making.
Customer	Registers on the platform, maintains a personal profile, searches available vehicles, views vehicle details, makes reservations, completes payments, downloads invoices, and tracks booking history. The customer interacts with the system throughout the vehicle rental lifecycle.	Quick and hassle-free vehicle booking, real-time availability information, transparent pricing, secure online payments, instant booking confirmations, easy access to invoices and rental history, and a smooth user experience across all platform features.

5. Functional Requirements

FR1 – User Registration

The system shall allow new customers to create an account by providing details such as name, email address, mobile number, password, address, and driving license information. The system shall validate the entered information, check for duplicate email addresses, and create a customer account upon successful registration.

FR2 – User Login Authentication

The system shall allow registered users to log in using their email address and password. The system shall verify the credentials, create a secure session, and grant access according to the assigned user role.

FR3 – Profile Management

The system shall allow customers to view and update their profile information, including contact details, address, and driving license information. Users shall also be able to change their password and manage their account details whenever required.

FR4 – Vehicle Search

The system shall allow customers to search for vehicles using filters such as category, location, rental price, fuel type, transmission type, and vehicle availability. The system shall display only the vehicles matching the selected search criteria.

FR5 – Vehicle Details

The system shall display complete details of a selected vehicle, including brand, model, category, specifications, rental charges, availability status, images, and vehicle description. Customers shall be able to review these details before making a booking.

FR6 – Vehicle Booking

The system shall allow customers to select a vehicle, choose pickup and return dates, specify rental locations, and submit a booking request. The system shall calculate the rental amount automatically and generate a unique booking reference number.

FR7 – Concurrent Booking Management

The system shall verify vehicle availability before confirming a booking request. It shall prevent multiple customers from booking the same vehicle for overlapping rental periods and update the availability status accordingly.

FR8 – Payment Processing

The system shall allow customers to make secure online payments for confirmed bookings. The system shall record transaction details, payment status, payment method, and payment date for future reference.

FR9 – Invoice Generation

The system shall automatically generate an invoice after successful payment processing. The invoice shall contain customer information, booking details, vehicle information, rental charges, taxes, and the final payable amount.

FR10 – Booking History

The system shall maintain a complete history of customer bookings. Customers shall be able to view active, completed, and cancelled bookings along with payment details, invoices, and booking status information.

FR11 – Vehicle Management

The system shall allow administrators to add, update, delete, and manage vehicle records. Administrators shall also be able to update rental charges, vehicle specifications, images, maintenance status, and availability information.

FR12 – Customer Management

The system shall allow administrators to view customer information, manage customer accounts, verify customer details, and access customer booking history whenever required.

FR13 – Booking Management

The system shall enable administrators to monitor, approve, modify, and cancel bookings. The system shall maintain the current status of each booking and record any updates made by the administrator.

FR14 – Payment Management

The system shall maintain records of all payment transactions and allow administrators to monitor payment status, transaction history, revenue details, and payment-related activities.

FR15 – Report Generation

The system shall generate reports related to bookings, customers, payments, revenue, vehicle utilization, and maintenance records. Administrators shall be able to view and download these reports for analysis and decision-making purposes.

FR16 – Admin Dashboard

The system shall provide a centralized dashboard displaying information such as total vehicles, active bookings, completed rentals, registered customers, revenue statistics, pending payments, and recent system activities.

FR17 – Notification System

The system shall send notifications regarding registration confirmation, booking confirmation, booking cancellation, payment updates, invoice generation, and vehicle return reminders through the configured communication channels.

FR18 – Secure Logout

The system shall allow users to securely log out of the application. Upon logout, the system shall terminate the active session and prevent unauthorized access to protected resources.

FR19 – Multi-Branch Management

The system shall allow administrators to create and manage multiple branch locations. Vehicles, bookings, customers, and rental records shall be associated with specific branches, enabling centralized management of operations across all locations.

6. Non-Functional Requirements

6.1 Performance

- The system shall respond to user requests within 2 seconds under normal operating conditions.
- The system shall support multiple users performing vehicle searches, bookings, and payments simultaneously without noticeable performance degradation.
- The system shall process booking confirmations, payment updates, and invoice generation efficiently.
- The system shall update vehicle availability immediately after a booking is confirmed.
- The system shall generate requested reports within 5 seconds under normal operating conditions.

6.2 Security

- The system shall provide secure user authentication using JWT-based authentication.
- User passwords shall be encrypted using bcrypt hashing before being stored in the database.
- The system shall implement Role-Based Access Control (RBAC) for Customers and Administrators.
- The system shall validate all user inputs to prevent unauthorized access and invalid data entry.
- The system shall allow access to sensitive information only to authorized users.

6.3 Reliability

- The system shall maintain at least 99% availability during normal operation.
- The system shall ensure accurate storage and retrieval of booking, payment, and customer records.
- The system shall prevent duplicate vehicle bookings through availability validation.
- Proper error handling mechanisms shall be implemented to prevent data loss during unexpected failures.

6.4 Scalability

- The system shall support at least 10 concurrent users without affecting core functionality.
- The architecture shall allow additional modules and features to be integrated with minimal code changes.
- The database shall efficiently handle growth in the number of customers, vehicles, bookings, payments, and branch locations.
- The system shall support expansion to multiple rental branches from a centralized platform.

6.5 Usability

- The application shall provide a simple and user-friendly interface for both customers and administrators.
- Navigation shall be clear and consistent across all pages.
- Customers shall be able to complete the vehicle booking process in a few straightforward steps.
- Error messages and validation messages shall be displayed clearly to guide users.

6.6 Maintainability

- The system shall follow a modular architecture using separate frontend, backend, and database layers.
- The codebase shall be structured, reusable, and easy to maintain.
- Updates, bug fixes, and feature enhancements shall be implemented without affecting existing modules.
- Proper documentation shall be maintained for future development and maintenance activities.

6.7 Portability

- The application shall run on Windows, Linux, and macOS development environments.
- Users shall be able to access the system through modern web browsers such as Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari.
- The system shall be deployable on local servers as well as cloud-based hosting platforms.

7. System Requirements

7.1 Hardware Requirements for developers

The minimum hardware requirements for developing and running the Vehicle Rental Booking Platform are:

- **Processor:** Intel Core i5 (or equivalent AMD processor)
- **RAM:** 8 GB or higher
- **Storage:** 256 GB SSD (minimum)
- **Internet Connection:** Required for development, testing, and deployment

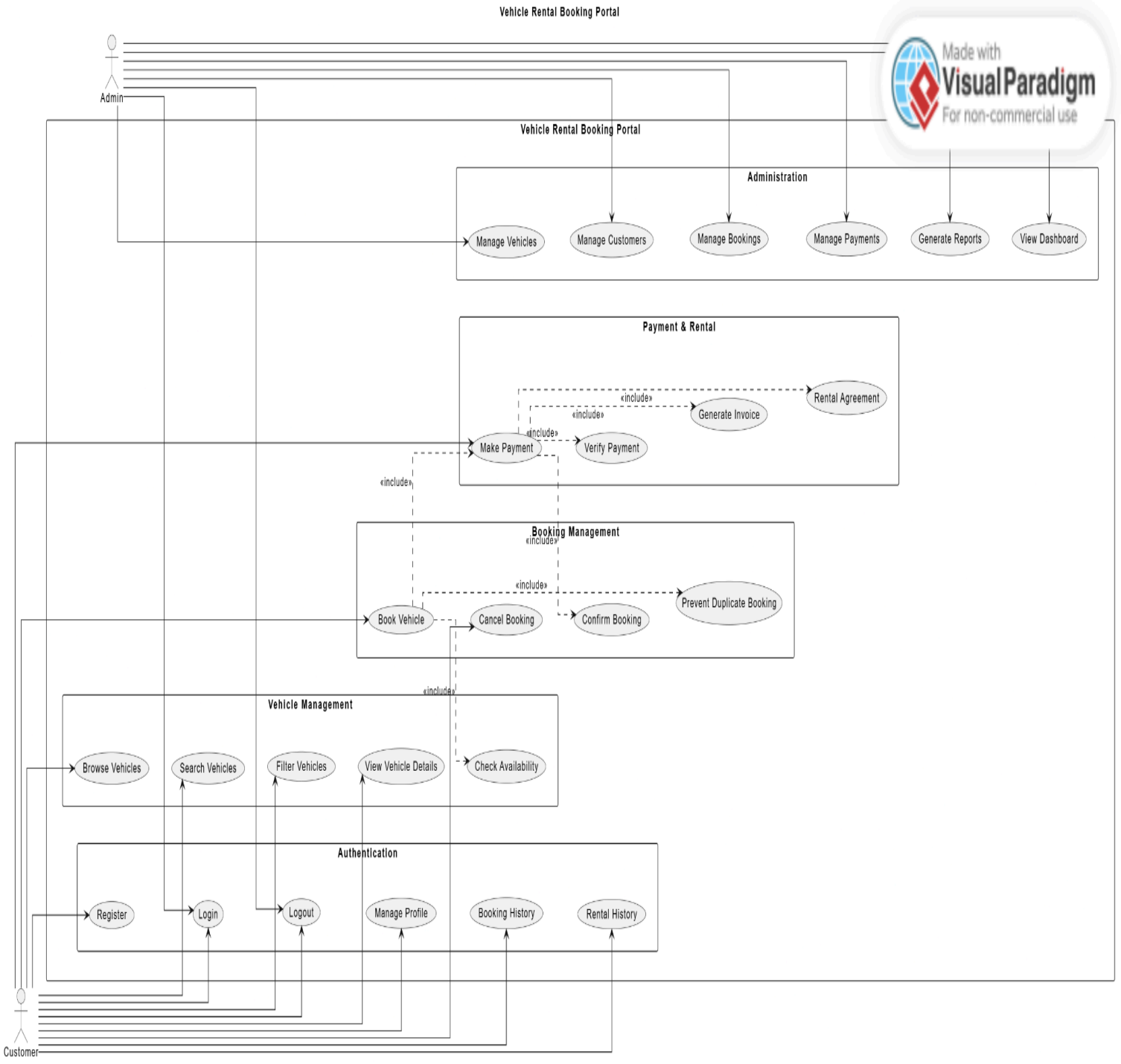
7.2 Software Requirements

The following software is required for the development and execution of the system:

- **Operating System:** Windows, Linux, or macOS
- **Code Editor:** Visual Studio Code
- **Frontend:** React.js, Vite, Tailwind CSS
- **Backend:** Node.js and Express.js
- **Database:** PostgreSQL

- **API Testing Tool:** Postman
- **Version Control:** Git and GitHub
- **Web Browser:** Google Chrome, Microsoft Edge, or Mozilla Firefox

8. Use Case Diagram



9. Use Case Descriptions

The Vehicle Rental Booking Portal Use Case Diagram represents the main interactions between the two actors—Customer and Admin—and the system's five major modules: Authentication, Vehicle Management, Booking Management, Payment & Rental, and Administration.

- Customer: Registers, logs in, manages profile, searches and views vehicles, books/cancels vehicles, makes payments, and views booking/rental history.
- Admin: Manages vehicles, customers, bookings, payments, reports, and the system dashboard.
- Module relationships: Booking includes availability checking, duplicate-booking prevention, confirmation, and payment; payment includes verification, invoice generation, and rental agreement generation.

Overall, the diagram shows who interacts with the system and which major functions each actor can perform.

10. Data Requirements

10.1 Database Tables

The Vehicle Rental Booking Platform uses the following database tables:

- **roles** – Stores system roles for access control.
- **users** – Stores user authentication and account information.
- **customer_profiles** – Stores customer personal, contact, identity, and address details.
- **vehicle_categories** – Stores vehicle categories and their characteristics.
- **branches** – Stores rental branch and location information.
- **vehicles** – Stores vehicle inventory and vehicle-specific details.
- **bookings** – Stores customer reservations and rental details.
- **payments** – Stores booking payment transactions.
- **invoices** – Stores invoices generated from payments.
- **rental_agreements** – Stores rental agreements associated with bookings.
- **reports** – Stores information about generated reports.
- **vehicle_images** – Stores images associated with vehicles.
- **booking_status_history** – Stores the status history of bookings.
- **vehicle_condition_media** – Stores before/after vehicle condition photos and videos.
- **maintenance_records** – Stores vehicle maintenance and service history.

- **audit_logs** – Records important administrative and system activities.

10.2 Important Attributes

TABLE	IMPORTANT ATTRIBUTES
roles	role_id, role_name, description, created_at, updated_at
users	user_id, role_id, email, password_hash, is_active, status, email_verified, last_login, created_at, updated_at, deleted_at
customer_profiles	customer_id, user_id, first_name, last_name, gender, date_of_birth, phone, alternate_phone, driving_license_no, license_expiry, aadhaar_number, address, city, state, country, postal_code, emergency_contact_name, emergency_contact_phone, profile_image, created_at, updated_at, deleted_at
vehicle_categories	category_id, category_name, description, seating_capacity, created_at, updated_at
branches	branch_id, branch_name, address, city, state, postal_code, phone, created_at, updated_at
vehicles	vehicle_id, category_id, branch_id, registration_number, brand, model, variant, manufacture_year, color, fuel_type, transmission, engine_capacity, seating_capacity, mileage, daily_rent, security_deposit, status, image_url, chassis_number, insurance_number, insurance_expiry, pollution_expiry, last_service_date, next_service_date, current_odometer, description, created_at, updated_at, deleted_at
bookings	booking_id, customer_id, vehicle_id, booking_reference, booking_date, pickup_date, return_date, actual_pickup, actual_return, pickup_location, return_location, booking_status, total_days, daily_rent, security_deposit, discount, tax, total_amount, cancellation_reason, cancelled_at, remarks, created_at, updated_at
payments	payment_id, booking_id, transaction_reference, payment_method, amount, payment_status, payment_gateway, gateway_response, payment_date, created_at, updated_at
invoices	invoice_id, payment_id, invoice_number, invoice_date, subtotal, tax_amount, total_amount, pdf_path, created_at, updated_at

rental_agreements	agreement_id, booking_id, agreement_number, agreement_date, agreement_status, document_path, signed_at, created_at, updated_at
reports	report_id, report_type, generated_by, generated_at, file_path, created_at
vehicle_images	image_id, vehicle_id, image_url, is_primary, uploaded_at
booking_status_history	history_id, booking_id, status, changed_by, changed_at, remarks
vehicle_condition_media	media_id, booking_id, uploaded_by, condition_stage, media_type, media_source, file_path, storage_provider, customer_visible, uploaded_at, description
maintenance_records	maintenance_id, vehicle_id, service_date, service_type, service_center, odometer_reading, service_cost, next_service_due, remarks, created_at
audit_logs	log_id, user_id, action, module, entity_id, ip_address, created_at

10.3 Relationships

- **roles → users (1:N)** – One role can be assigned to multiple users.
- **users → customer_profiles (1:1)** – One user can have one customer profile.
- **vehicle_categories → vehicles (1:N)** – One category can contain multiple vehicles.
- **branches → vehicles (1:N)** – One branch can have multiple vehicles.
- **customer_profiles → bookings (1:N)** – One customer can create multiple bookings.
- **vehicles → bookings (1:N)** – One vehicle can be associated with multiple bookings over time.
- **bookings → payments (1:1)** – Each booking has one payment record in the current design.
- **payments → invoices (1:1)** – Each payment can have one invoice.
- **bookings → rental_agreements (1:1)** – Each booking can have one rental agreement.
- **users → reports (1:N)** – One user can generate multiple reports.
- **vehicles → vehicle_images (1:N)** – One vehicle can have multiple images.
- **bookings → booking_status_history (1:N)** – One booking can have multiple status-history records.
- **users → booking_status_history (1:N)** – One user can make multiple booking-status changes.

- **bookings → vehicle_condition_media (1:N)** – One booking can have multiple condition photos/videos.
- **users → vehicle_condition_media (1:N)** – One user can upload multiple condition media files.
- **vehicles → maintenance_records (1:N)** – One vehicle can have multiple maintenance records.
- **users → audit_logs (1:N)** – One user can generate multiple audit-log entries.

11. Assumptions and Constraints

11.1 Assumptions

- Users have a stable internet connection to access the application.
- Users possess basic knowledge of using a web application.
- The required hardware and software are available for development and deployment.
- The database and server remain available during normal system operation.
- Valid user information will be provided during registration and booking.

11.2 Constraints

- The project will be developed within the given academic timeline (approximately 12 weeks).
- Development will be carried out with limited resources and budget.
- The system is designed as a web application and does not include a mobile app in the current version.
- Third-party services such as payment gateways may require additional configuration during deployment.
- The project scope is limited to the features defined in the Software Requirements Specification.

12. Future Enhancements

The current version of the Vehicle Rental Booking Platform covers the essential features required for managing vehicle rentals. In the future, the system can be enhanced by adding the following features:

- **AI-Based Recommendations:** Suggest suitable vehicles to customers based on their booking history and preferences.
- **Mobile Application:** Develop Android and iOS applications for easier access and booking on mobile devices.
- **AI Chatbot:** Integrate a chatbot to answer customer queries, assist with bookings, and provide 24/7 support.
- **Advanced Analytics Dashboard:** Display detailed insights on bookings, revenue, vehicle utilization, and customer trends through interactive charts.
- **Live GPS Tracking:** Allow customers and administrators to track rented vehicles in real time.
- **Multiple Payment Gateways:** Support additional payment options such as UPI, digital wallets, and international payment methods.
- **Email and SMS Notifications:** Send automatic booking confirmations, payment receipts, and rental reminders.

13. References

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APIs / Libraries

- JWT (JSON Web Token)
- bcrypt
- Axios
- React Router DOM
- React Hook Form